

LABSHEET - 1**HIMANI CHAUDHARY****BTECH(CSE) 3RD YEAR****CU240250967****SECTION - B****ROLL NO. - 24****Mastery in Data Analytics and Visualizations and Career Advancement (Technical)**

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
import os
```

```
# 1. LOAD IMAGE
# =====

image_path = "/content/modern-business-building-with-glass-wall-from-empty-floor.jpg"
image = cv2.imread(image_path)

if image is None:
    print("ERROR: Image not found!")
    print("Check that the image path and filename are correct.")
    exit()

print("Image loaded successfully!")
print("Himani Chaudhary")
print("CU240250967")
```

```
Image loaded successfully!
Himani Chaudhary
CU240250967
```

```
# 2. DISPLAY ORIGINAL IMAGE
# =====

import cv2
from google.colab.patches import cv2_imshow
import matplotlib.pyplot as plt

# Read the image
image = cv2.imread("/content/modern-business-building-with-glass-wall-from-empty-floor.jpg")

# Display using Colab's OpenCV-compatible function
cv2_imshow(image)

# Convert BGR to RGB for Matplotlib
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# Display using Matplotlib
plt.imshow(image_rgb)
plt.axis("off")
plt.show()

print("Himani Chaudhary")
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```



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```
# 3. IMAGE PROPERTIES
# =====

height, width, channels = image.shape

print("\n===== IMAGE PROPERTIES =====")
print("Width      :", width)
print("Height     :", height)
print("Channels    :", channels)
print("Resolution  :", width, "x", height)
print("Data Type   :", image.dtype)
print("Total Pixels:", image.size)
print("Himani Chaudhary")
print("CU240250967")
```

```
===== IMAGE PROPERTIES =====  
Width      : 5000  
Height     : 4026  
Channels   : 3  
Resolution : 5000 x 4026  
Data Type  : uint8  
Total Pixels: 60390000  
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```

```
# 4. SAVE IMAGE IN JPEG AND PNG  
# =====  
  
cv2.imwrite("output_image.jpg", image)  
cv2.imwrite("output_image.png", image)  
  
print("\nImages saved successfully:")  
print("output_image.jpg")  
print("output_image.png")  
print("Himani Chaudhary")  
print("CU240250967")
```

Images saved successfully:
output_image.jpg
output_image.png
Himani Chaudhary
CU240250967

```
# 5. CONVERT TO GRAYSCALE  
# =====  
  
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)  
  
cv2.imwrite("grayscale.jpg", gray)  
print("Himani Chaudhary")  
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```

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```
# 6. CONVERT TO HSV  
# =====  
  
hsv = cv2.cvtColor(image, cv2.COLOR_BGR2HSV)  
  
cv2.imwrite("hsv.jpg", hsv)  
print("Himani Chaudhary")  
print("CU240250967")
```

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```
# 7. CONVERT TO LAB  
# =====  
  
lab = cv2.cvtColor(image, cv2.COLOR_BGR2LAB)  
  
cv2.imwrite("lab.jpg", lab)  
print("Himani Chaudhary")  
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```

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```
# 8. RESIZE IMAGE  
# =====  
  
new_width = 400  
new_height = 300  
  
resized = cv2.resize(image, (new_width, new_height))  
  
cv2.imwrite("resized.jpg", resized)  
print("Himani Chaudhary")  
print("CU240250967")
```

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```
# 9. ROTATE IMAGE
# =====

# Rotate 90 degrees clockwise
rotated = cv2.rotate(image, cv2.ROTATE_90_CLOCKWISE)

cv2.imwrite("rotated.jpg", rotated)
print("Himani Chaudhary")
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```

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```
# 10. HORIZONTAL FLIP
# =====

horizontal_flip = cv2.flip(image, 1)

cv2.imwrite("horizontal_flip.jpg", horizontal_flip)
print("Himani Chaudhary")
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```

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```
# 11. VERTICAL FLIP
# =====

vertical_flip = cv2.flip(image, 0)

cv2.imwrite("vertical_flip.jpg", vertical_flip)
print("Himani Chaudhary")
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```

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```
# 12. IMAGE COMPLEMENT / NEGATIVE
# =====

negative = 255 - image

cv2.imwrite("negative.jpg", negative)
print("Himani Chaudhary")
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```

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```
# 13. CROP REGION OF INTEREST (ROI)
# =====

h, w = image.shape[:2]

# Select the center 50% of the image
x1 = int(w * 0.25)
x2 = int(w * 0.75)

y1 = int(h * 0.25)
y2 = int(h * 0.75)

roi = image[y1:y2, x1:x2]

cv2.imwrite("roi.jpg", roi)
print("Himani Chaudhary")
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```

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```
# 14. DISPLAY ALL PROCESSED IMAGES
# =====

plt.figure(figsize=(14, 10))

# Original
plt.subplot(3, 3, 1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title("Original")
plt.axis("off")

# Grayscale
plt.subplot(3, 3, 2)
plt.imshow(gray, cmap="gray")
plt.title("Grayscale")
plt.axis("off")

# HSV
plt.subplot(3, 3, 3)
plt.imshow(cv2.cvtColor(hsv, cv2.COLOR_HSV2RGB))
plt.title("HSV")
plt.axis("off")

# LAB
plt.subplot(3, 3, 4)
plt.imshow(cv2.cvtColor(lab, cv2.COLOR_LAB2RGB))
plt.title("LAB")
plt.axis("off")

# Resized
plt.subplot(3, 3, 5)
plt.imshow(cv2.cvtColor(resized, cv2.COLOR_BGR2RGB))
plt.title("Resized")
plt.axis("off")

# Rotated
plt.subplot(3, 3, 6)
plt.imshow(cv2.cvtColor(rotated, cv2.COLOR_BGR2RGB))
plt.title("Rotated")
plt.axis("off")

# Horizontal Flip
plt.subplot(3, 3, 7)
plt.imshow(cv2.cvtColor(horizontal_flip, cv2.COLOR_BGR2RGB))
plt.title("Horizontal Flip")
plt.axis("off")

# Negative
plt.subplot(3, 3, 8)
plt.imshow(cv2.cvtColor(negative, cv2.COLOR_BGR2RGB))
plt.title("Negative")
plt.axis("off")

# ROI
plt.subplot(3, 3, 9)
plt.imshow(cv2.cvtColor(roi, cv2.COLOR_BGR2RGB))
plt.title("ROI")
plt.axis("off")

plt.tight_layout()
plt.show()
print("Himani Chaudhary")
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```



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```
# 15. COMPLETION MESSAGE
# =====

print("\n=====")
print("ALL OPERATIONS COMPLETED!")
print("=====")

print("\nOutput files created:")
print("1. output_image.jpg")
print("2. output_image.png")
print("3. grayscale.jpg")
print("4. hsv.jpg")
print("5. lab.jpg")
print("6. resized.jpg")
print("7. rotated.jpg")
print("8. horizontal_flip.jpg")
print("9. vertical_flip.jpg")
print("10. negative.jpg")
print("11. roi.jpg")

print("\nPress any key on an image window to close.")
print("Himani Chaudhary")
print("CU240250967")
```

```
=====
ALL OPERATIONS COMPLETED!
=====
```

```
Output files created:
1. output_image.jpg
2. output_image.png
3. grayscale.jpg
4. hsv.jpg
5. lab.jpg
6. resized.jpg
7. rotated.jpg
```